

jcm-strat Phase 6: a stratosphere without radiation

What was tried, why, and what came out. Eight one-year runs of 2005, all on the Phase 3 model (ERA5-nudged troposphere below 150 hPa, four passive tracers, T63L95, 12 minute step), differing only in how the stratospheric temperature is held. Record as of 3 September 2026, branch phase6-stratosphere.

1. The problem

Phases 1 to 4 held the stratosphere with Held-Suarez relaxation. Its target temperature is clamped at 200 K everywhere above the tropopause, and the model is pulled toward that target with a 40 day timescale. The result is an isothermal stratosphere with no polar-night jet and no seasons, 35 K RMS away from ERA5. Radiation would fix this but is the whole cost of the full model, so Phase 6 asks: can an analytic target do it instead?

2. The recipe: Polvani-Kushner (2002) with three added knobs

Polvani and Kushner replace the 200 K clamp above 100 hPa with a blend of two temperature profiles. Outside the winter polar cap the target is the US Standard Atmosphere, which is about 217 K at 100 hPa and warms slowly with height. Inside the winter cap it is a 'polar vortex' profile that starts at the same 217 K and cools with height at a fixed rate γ . The colder the winter pole relative to midlatitudes, the stronger the jet that thermal-wind balance demands. The cap edge is at 50 degrees latitude with a 10 degree width. Polvani-Kushner was designed as a perpetual-winter setup; here the winter hemisphere follows the calendar. Four knobs were varied:

knob	what it does	values tried
gamma	vortex cooling rate with height inside the winter cap, K per km. Sets how cold the winter pole is and hence how strong the jet is.	4 (Polvani-Kushner's strong case), 2
tau	relaxation time in the stratosphere: how fast the model temperature is pulled toward the target. Held-Suarez and Polvani-Kushner use 40 days; the real radiative damping time is 5 to 20 days.	40, 25, 15 days
season	how the cooling amplitude varies through the year. The first version was a half-year cosine peaking mid-January; the wide version starts at the autumn equinox, holds through winter and ends after the spring equinox.	narrow (cosine), wide (offset 0.5)
taper	the pressure above which the vortex cooling is faded out. Polvani-Kushner's profile cools without limit, asking for 143 K at 3 hPa where the real winter stratopause is warm.	none, 10, 5, 3 hPa

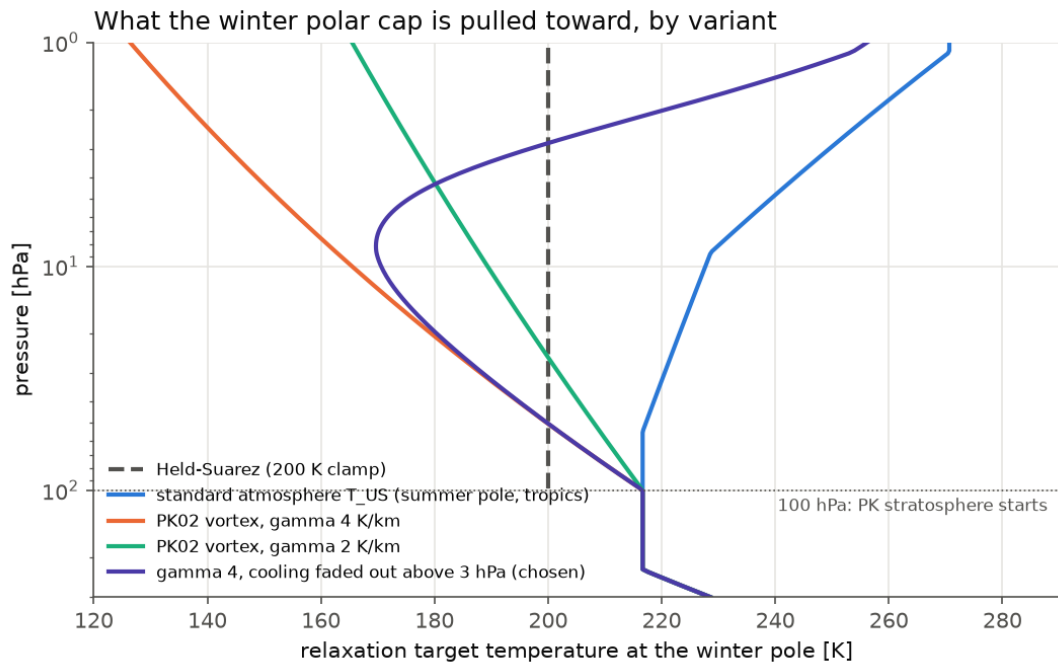


Figure 1. The temperature the winter polar cap is pulled toward, for each variant. Held-Suarez is a flat 200 K (dashed). The standard atmosphere (blue) is what the summer pole and the tropics see in every Polvani-Kushner run. Gamma 4 (orange) and gamma 2 (aqua) keep cooling with height; the chosen variant (violet) follows gamma 4 down to about 10 hPa and then returns to the standard atmosphere above 3 hPa.

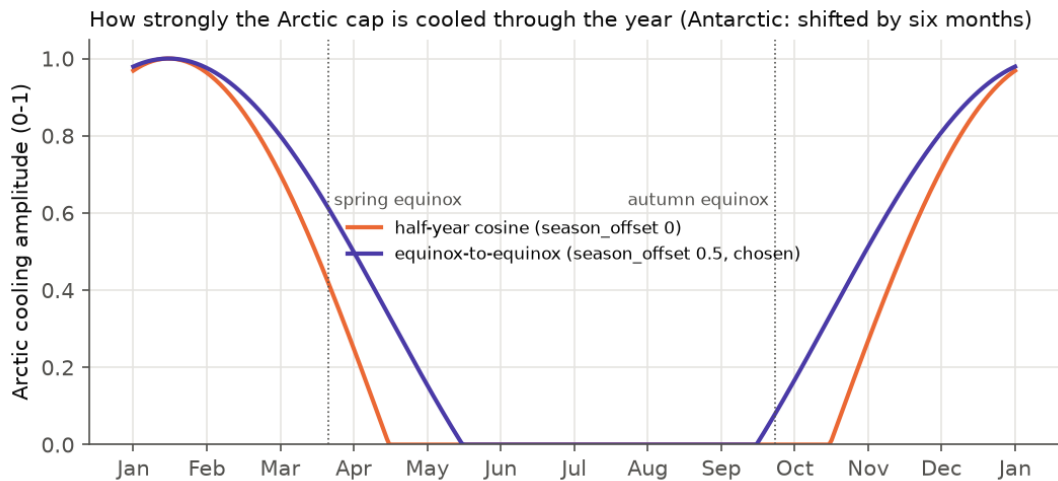


Figure 2. The seasonal envelope. With the narrow cosine the Arctic cap is cooled at only 7 percent of full strength in mid-October, when ERA5 has already cooled 15 K. The wide envelope starts cooling at the autumn equinox.

3. The runs, in the order they were tried

Each row changes one thing relative to an earlier row. All numbers are for 2005, against ERA5, over 100 to 1 hPa. 'u 60N DJF' is the Arctic polar-night jet at 10 hPa in winter; ERA5 has 38 m/s. 'u 60S JJA' is the Antarctic jet; ERA5 has 78 m/s.

run	gamma	tau	season	taper	T RMSE [K]	u RMSE [m/s]	u 60N DJF	u 60S JJA	what it tested
Held-Suarez (Phase 3)	-	40	-	-	35.2	13.0	4	10	the baseline
PK02 as published	4	40	narrow	none	8.9	10.2	22	41	does the analytic vortex work at all
gamma 2	2	40	narrow	none	7.8	10.1	14	28	is a weaker vortex closer to reality: no
wide season	4	40	wide	none	10.1	9.5	24	40	timing of onset and breakdown
tau 15	4	15	narrow	none	7.4	7.7	40	65	radiative damping timescale
tau 25, wide	4	25	wide	none	9.2	7.5	27	58	is the response to tau monotonic: yes
tau 15, wide	4	15	wide	none	8.5	8.4	43	68	combine the two improvements
taper 10 hPa	4	15	wide	10	6.4	7.6	25	54	cap the cooling aloft; too low, kills the jet
taper 5 hPa	4	15	wide	5	6.5	6.9	34	62	intermediate
taper 3 hPa (chosen)	4	15	wide	3	6.7	6.8	38	66	best winds, temperature nearly as good

Reading the table

- **Gamma:** halving it weakens both jets and gains nothing elsewhere. Gamma 4 stays.
- **Tau is the dominant knob.** Going from 40 to 15 days halves the wind error and takes both jets from about 55 percent of ERA5 to within 15 percent. The intermediate value 25 lands in between on every metric.
- **The wide season** barely moves the RMSE but fixes timing: the Arctic cooling starts a month earlier and the Antarctic vortex holds until November instead of collapsing in mid-October (Figure 3).
- **The taper** is a trade: 10 hPa gives the best temperature but weakens the 10 hPa jet by 20 m/s; 3 hPa keeps the jet and still removes the cold pole aloft. It was chosen.
- **The DJF temperature numbers are inflated** for every run: the ERA5 initial state has no information above 50 hPa, so January is spent spinning the upper stratosphere up. The annual column is the fair one. The multi-year chain removes this.

4. What the plots show

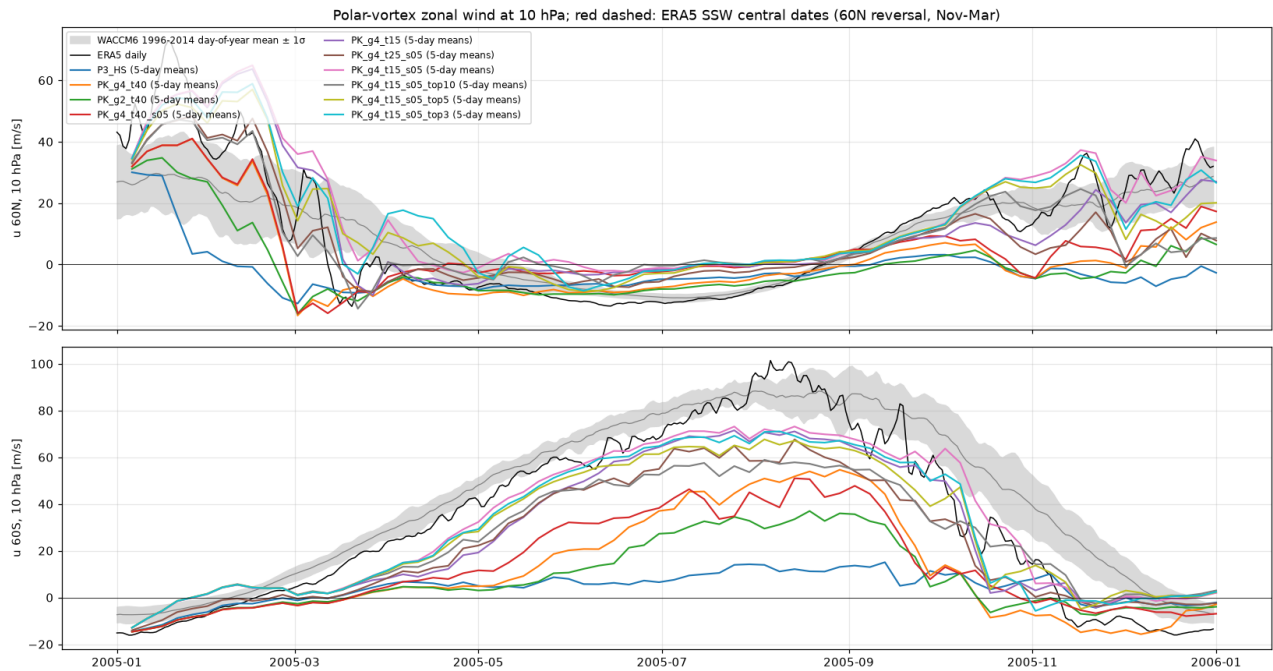


Figure 3. Zonal wind at 10 hPa at 60N (top) and 60S (bottom) through 2005. Black: ERA5 daily. Grey band: WACCM6 day-of-year mean and spread over 1996 to 2014. Held-Suarez (dark blue) has no vortex in either hemisphere. The tau 40 runs (orange, green, red) reach half of ERA5's Antarctic jet and collapse in mid-October. The tau 15 runs (purple, pink, cyan) reach 65 to 70 m/s and, with the wide season, persist to November. The 10 hPa taper (grey) sits visibly below the other tau 15 runs.

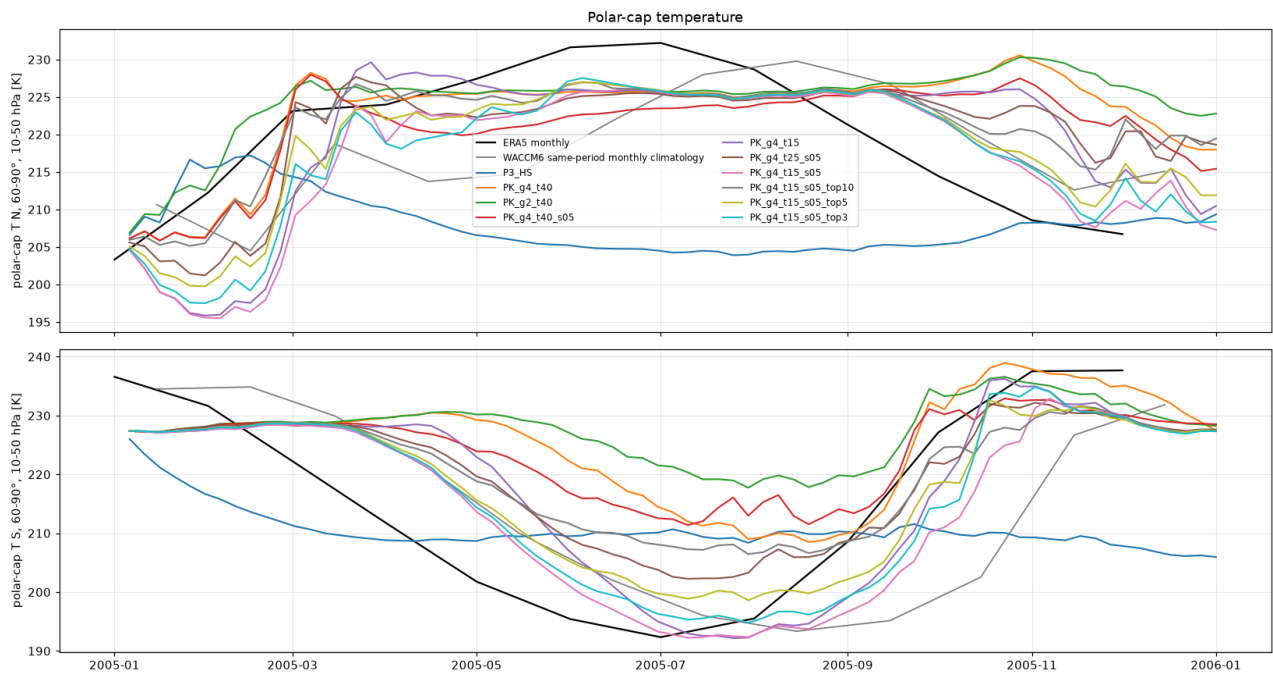


Figure 4. Polar-cap temperature, 60 to 90 degrees, averaged over 10 to 50 hPa. Black: ERA5 monthly. Grey: WACCM6 same-period climatology. The tau 40 runs never get colder than 210 K in the Antarctic winter; the tau 15 runs reach ERA5's 192 K and follow its warming in October. In the Arctic the tau 15 runs overshoot ERA5's February by about 10 K, which is why the NH jet in those runs is at the strong end.

5. Before and after: the full latitude-height picture

Four columns: temperature and zonal wind for December to February, then for June to August. Figure 5 puts every model version on one page: JCM's full ECHAM physics under the same nudging, Held-Suarez (Phases 1 to 4), the chosen Polvani-Kushner configuration, then the two references ERA5 and WACCM6. The full-physics run has the temperatures but almost no Arctic vortex; Held-Suarez has neither; Polvani-Kushner has both jets at 80 to 100 percent of ERA5's strength.

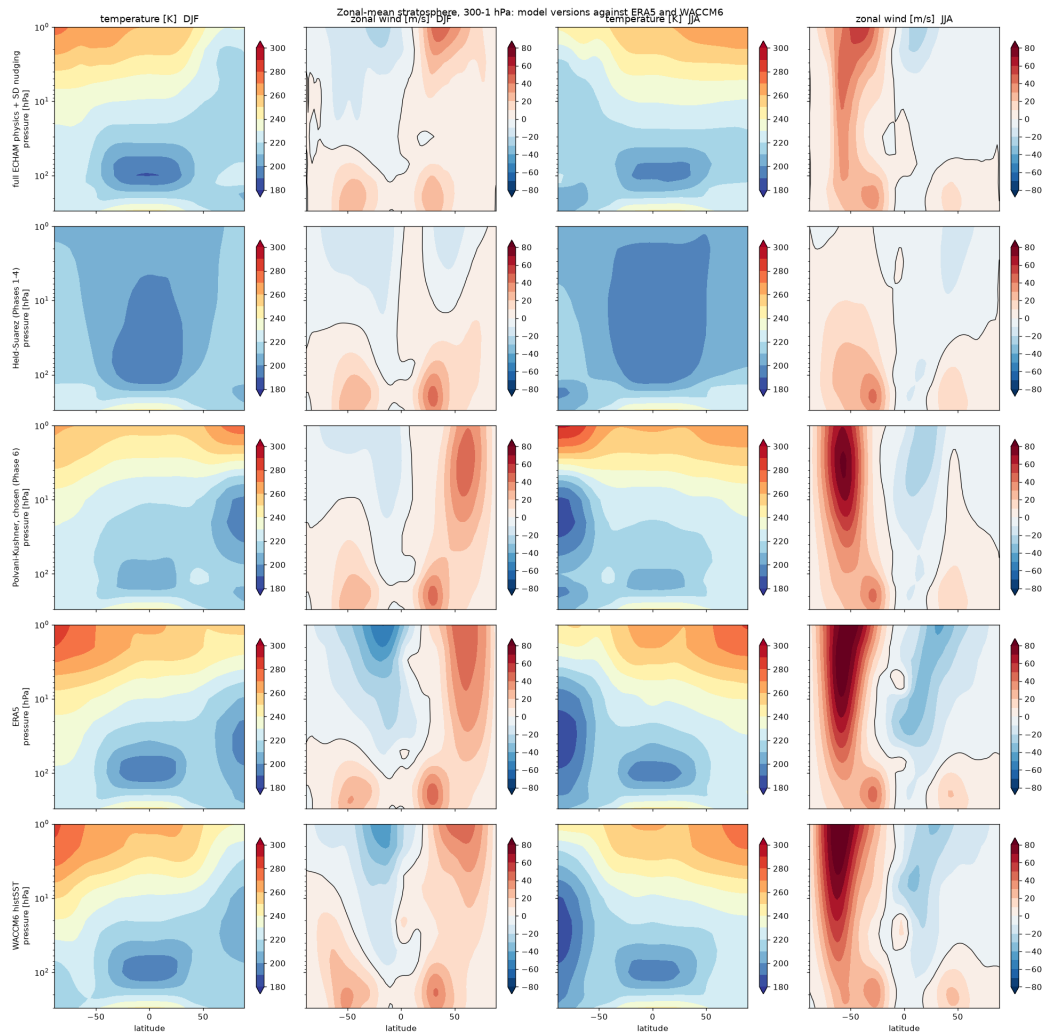


Figure 5. Zonal-mean temperature and zonal wind, 300 to 1 hPa, DJF and JJA, for 2005. Rows: full ECHAM physics with the same ERA5 nudging below 150 hPa, Held-Suarez, the chosen Polvani-Kushner run, ERA5, WACCM6.

The next three figures repeat Held-Suarez, Polvani-Kushner as published and the chosen run individually, each with ERA5, WACCM6 and the model-minus-ERA5 difference as the bottom row. Blue in the bottom row means the model is too cold or too westerly-weak.

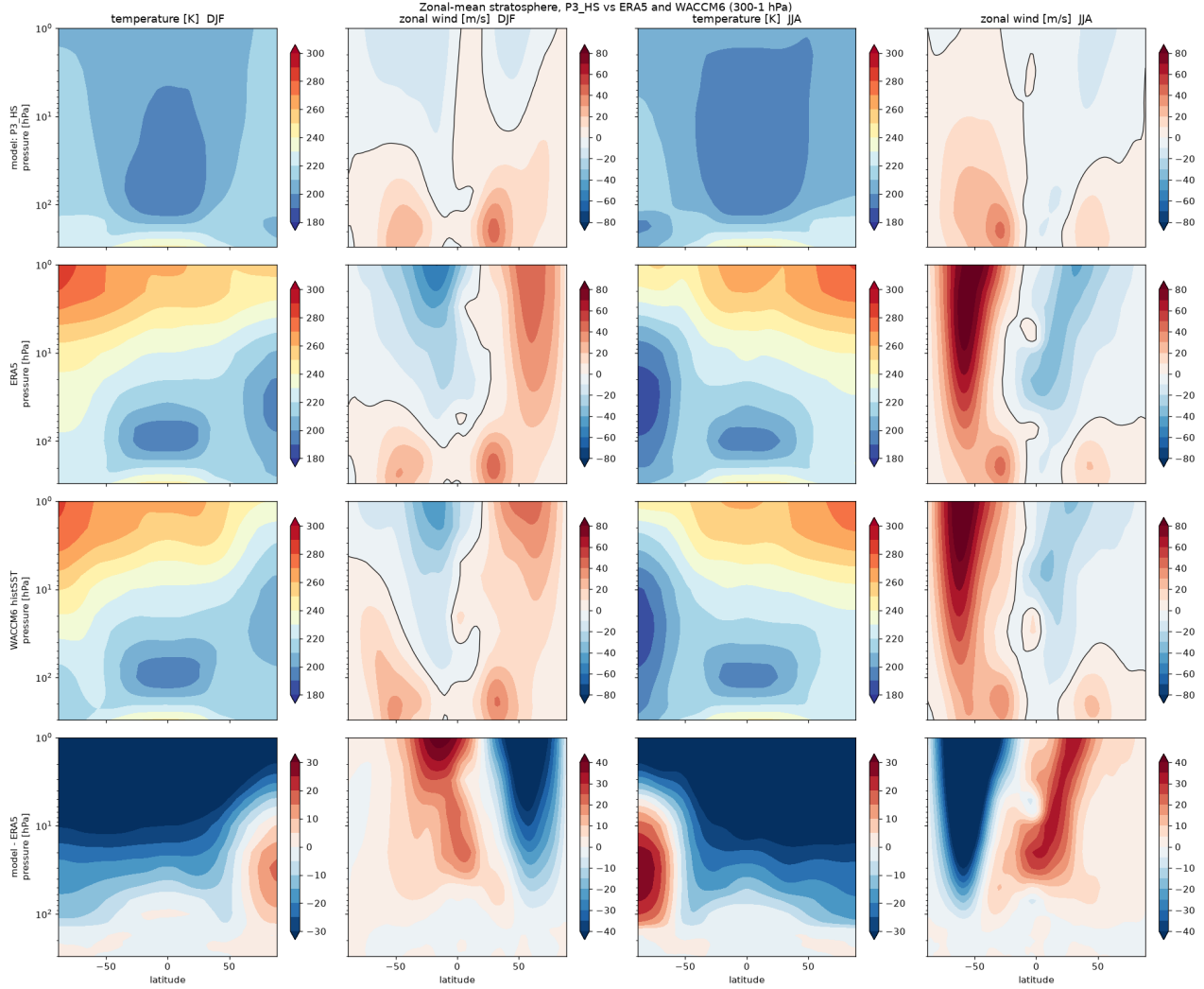


Figure 5a. Held-Suarez (Phase 3). A uniform 200 K stratosphere, no jets above 100 hPa, 30 K too cold nearly everywhere.

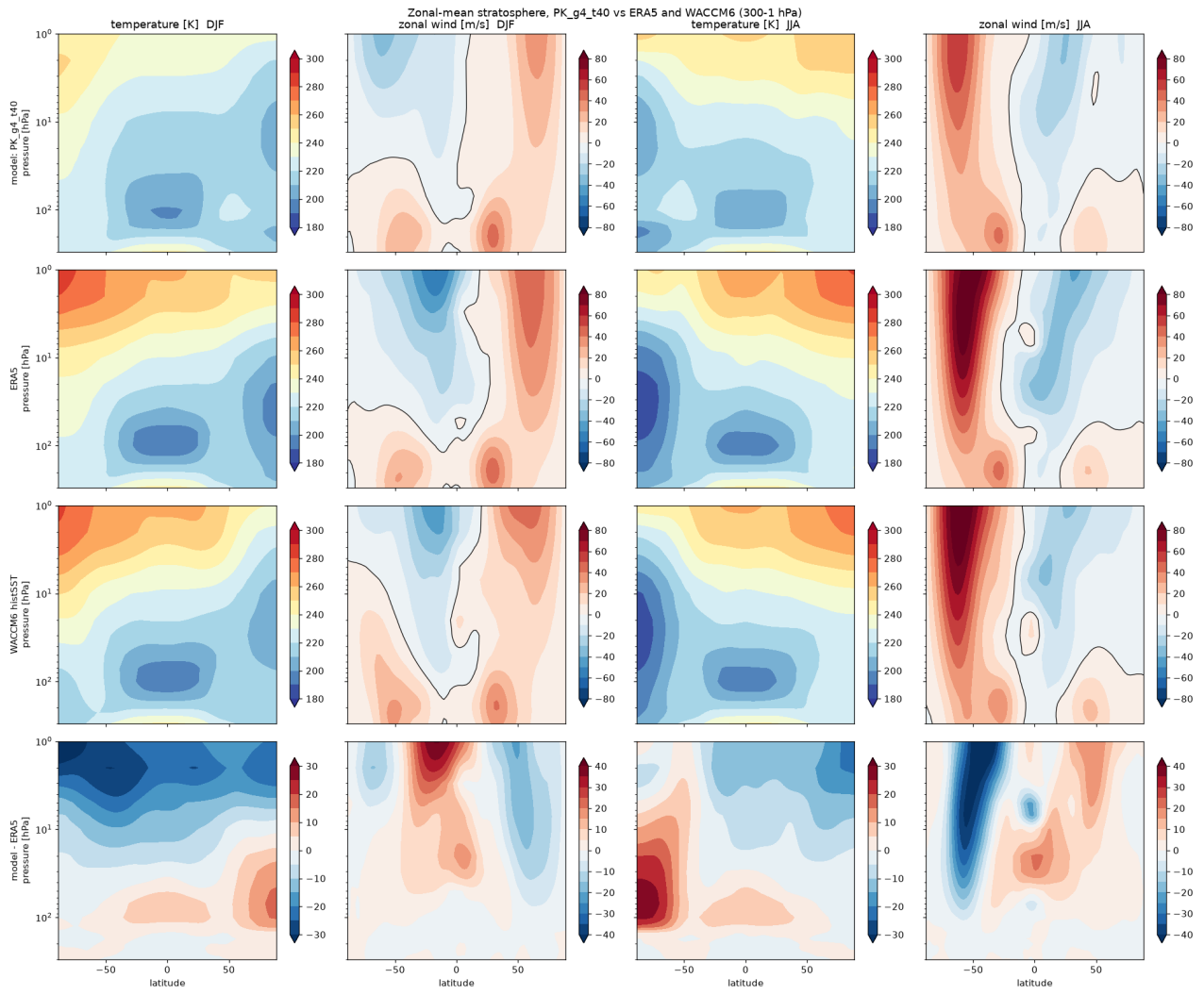


Figure 6. Polvani-Kushner as published (gamma 4, tau 40). Both jets appear, in the right place but at half strength. The upper stratosphere is 20 to 30 K too cold in DJF and the tropical lower stratosphere 10 K too warm.

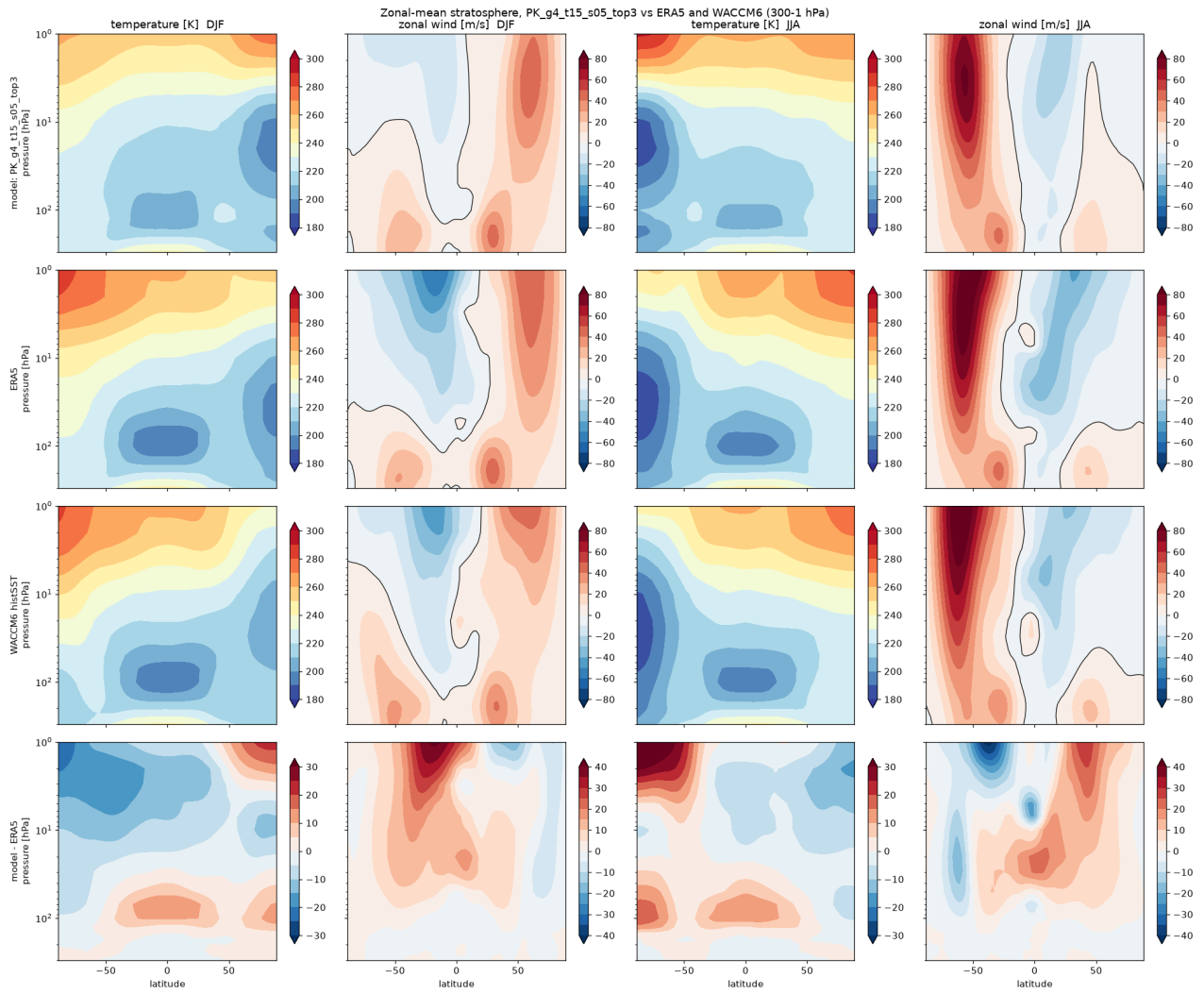


Figure 7. The chosen configuration (gamma 4, tau 15, wide season, taper 3 hPa). The Antarctic jet reaches 70 m/s against ERA5's 80 to 90, the Arctic jet matches, the winter poles are within 10 K. What remains: a 9 K cold layer at 1 to 3 hPa year-round, and the 10 K warm tropical lower stratosphere, both properties of the standard-atmosphere target.

6. Cost, decision and what is running

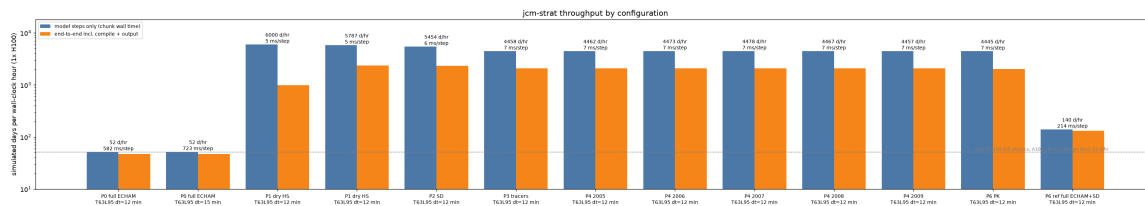


Figure 8. Simulated days per wall-clock hour on one H100. The Polvani-Kushner term adds nothing measurable to the Phase 3 cost.

Decision

The chosen settings, gamma 4 K/km, tau 15 days, wide season, taper at 3 hPa, are now the defaults of the strat_pk physics, so a plain run of the p6_pk experiment means this configuration. Polvani-Kushner as published is recoverable by setting tau 40, season offset 0 and no taper.

Open points carried forward

- A short tau also damps internal variability: none of the tau 15 runs produces a mid-winter wind reversal in 2005. That is correct for 2005, which had no sudden stratospheric warming, but 2006, 2008 and 2009 did. The five-year chain now running with the chosen configuration is the test; if the warmings are suppressed, tau 25 is the fallback.
- The tropical lower stratosphere is 10 K too warm because the standard atmosphere has no tropical tropopause. A latitude-dependent floor is the obvious fix and matters for the tropical pipe and age of air.
- The 1 to 3 hPa layer is 9 K too cold year-round even at tau 15, so something dynamical is cooling it. Gravity-wave drag and RRTMGP are the two things that could tell.

Against JCM's own full physics

The same 2005 year was run with the complete ECHAM package (RRTMGP radiation, convection, clouds, boundary-layer diffusion, surface scheme, both gravity-wave drags) under the same ERA5 nudging below 150 hPa, as the reference for what radiation buys. It has the better temperature climatology, 4.3 K RMSE against 6.7 K for the analytic stratosphere, but almost no Arctic vortex: the 10 hPa wind at 60N falls to about 4 m/s in the first week of January and never rebuilds in November and December, and the Antarctic jet peaks at 40 to 50 m/s against ERA5's 80 to 100. Its wind RMSE is 9.7 m/s against 6.8 m/s. It also runs 32 times slower. Caveats: its nudging target had to be 12-hourly for memory reasons, and it is one year.

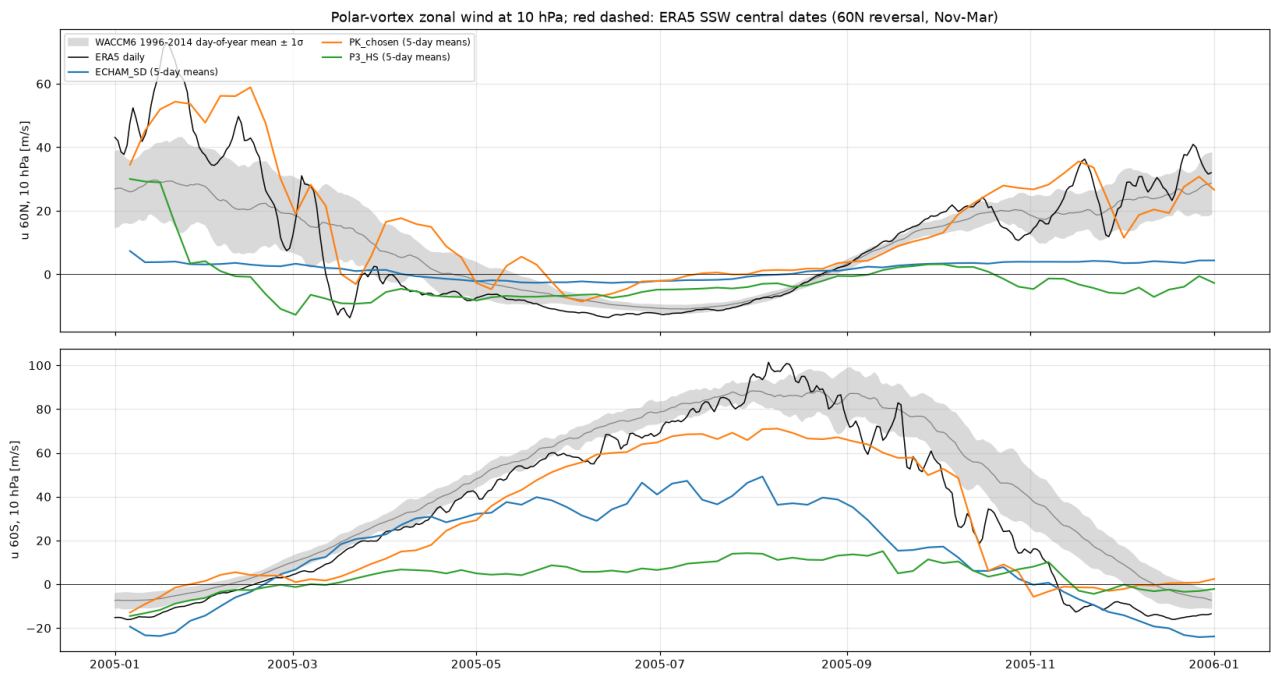


Figure 8b. The 10 hPa vortex wind in 2005: full ECHAM (blue), the chosen analytic stratosphere (orange), Held-Suarez (green), ERA5 (black) and the WACCM6 envelope (grey). The full-physics run has no Arctic vortex at all after the first week and half an Antarctic one.

Reference data

ERA5 monthly u and T on 25 levels to 1 hPa and daily u at 10 hPa for 2005 to 2009 from the Copernicus Climate Data Store; WACCM6 from the CESM2.1.5 histSST member 1996 to 2014 on disk (monthly T and U , daily zonal-mean U), used as a same-period climatology because its dynamics are free-running. Details in docs/outputs/06_stratosphere/output.md.

7. Five years with the chosen configuration, 2005 to 2009

The chosen settings were run as five chained years with the four passive tracers (about 65 minutes on GPU 0) and compared with ERA5 for the same years, with WACCM6, with CLaMS age of air, and with the Held-Suarez five-year chain from Phase 4.

metric, 2005 to 2009	Held-Suarez chain	Polvani-Kushner chain	reference
T RMSE 100 to 1 hPa vs ERA5	35.7 K	6.4 K	
u RMSE 100 to 1 hPa vs ERA5	11.9 m/s	5.6 m/s	
u at 60N, 10 hPa, DJF	-2 m/s	32 m/s	ERA5 28 m/s
u at 60S, 10 hPa, JJA	8 m/s	65 m/s	ERA5 72 m/s
SSW winters reproduced (2006, 2008, 2009)	0 of 3	2 of 3 (2008, 2009)	ERA5 daily
age-of-air contrast tropics vs 50 to 70 deg, 20 km	1.21 yr	1.56 yr	CLaMS 2.79 yr
age-of-air contrast, 30 km	0.35 yr	0.77 yr	CLaMS 0.88 yr
tracer conservation (unity, sai, minima)	pass	pass	

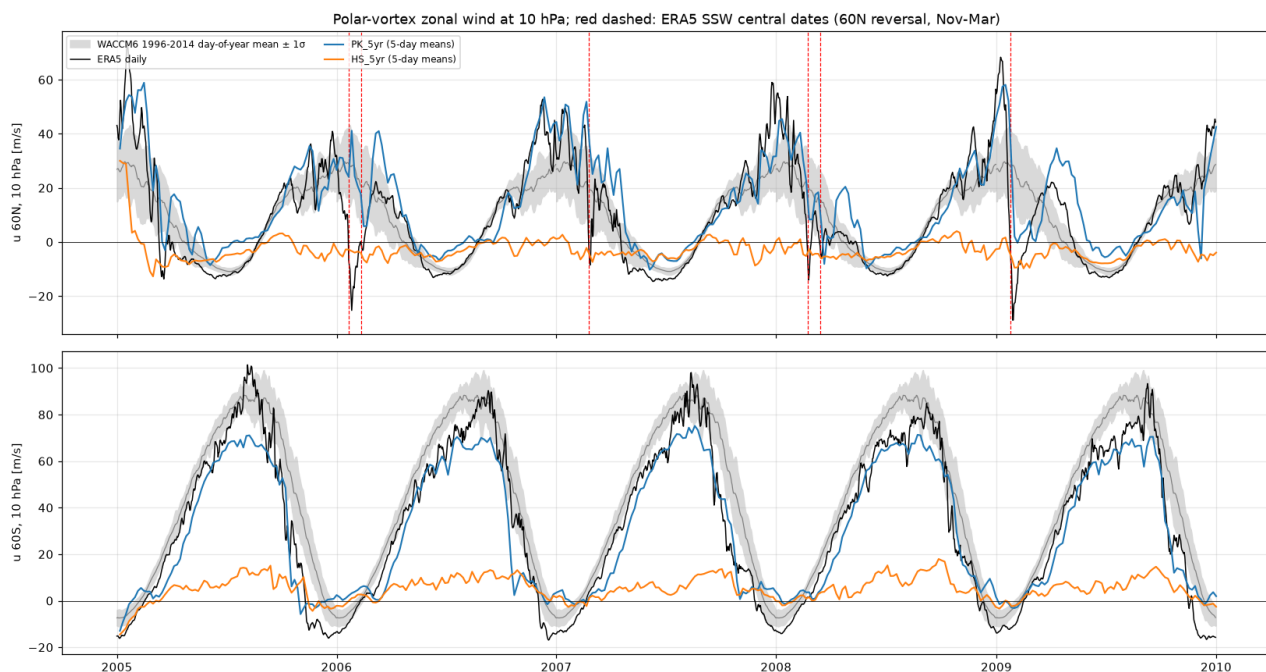


Figure 9. Five winters of the 10 hPa vortex wind. Black: ERA5 daily; red dashed: ERA5 sudden-warming dates. Blue: the Polvani-Kushner chain. Orange: the Held-Suarez chain from Phase 4. The Arctic vortex now varies from year to year and reverses in the 2008 and 2009 warmings; 2006 and 2007 are missed. The Antarctic vortex reaches 70 m/s each winter against 80 to 100 in ERA5.

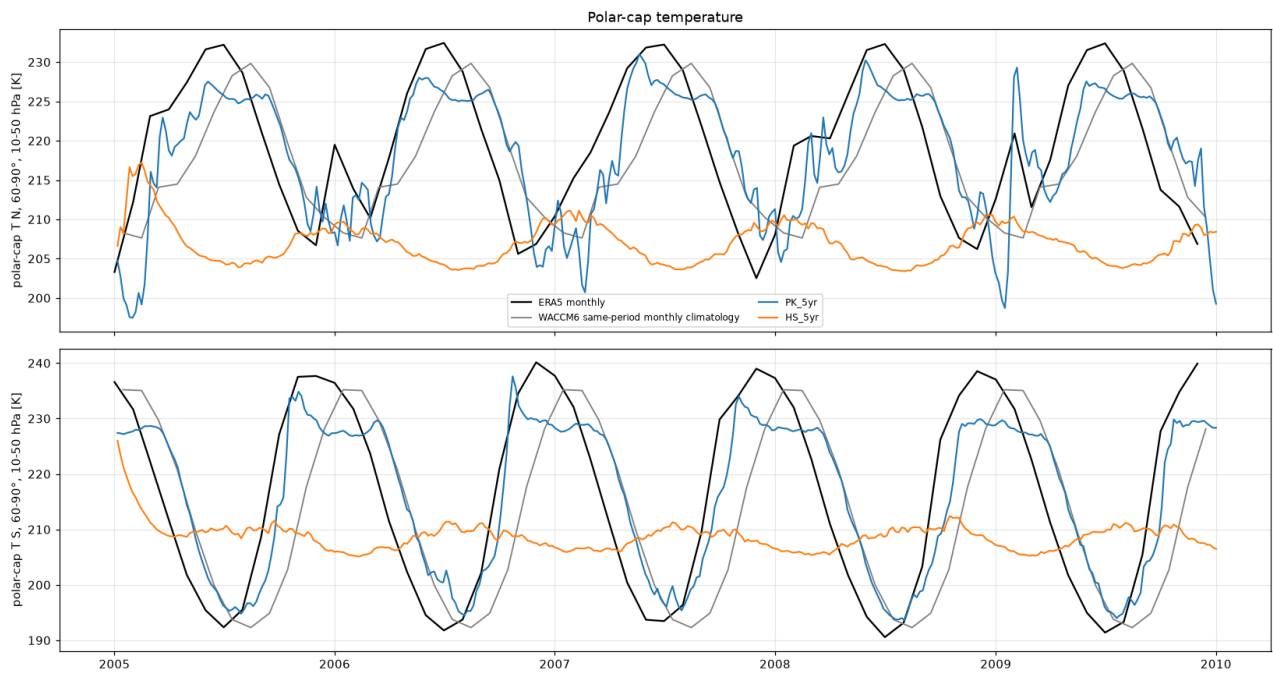


Figure 10. Polar-cap temperature over five years. The Polvani-Kushner chain (blue) follows ERA5 (black) in both hemispheres; Held-Suarez (orange) has no seasonal cycle at all.

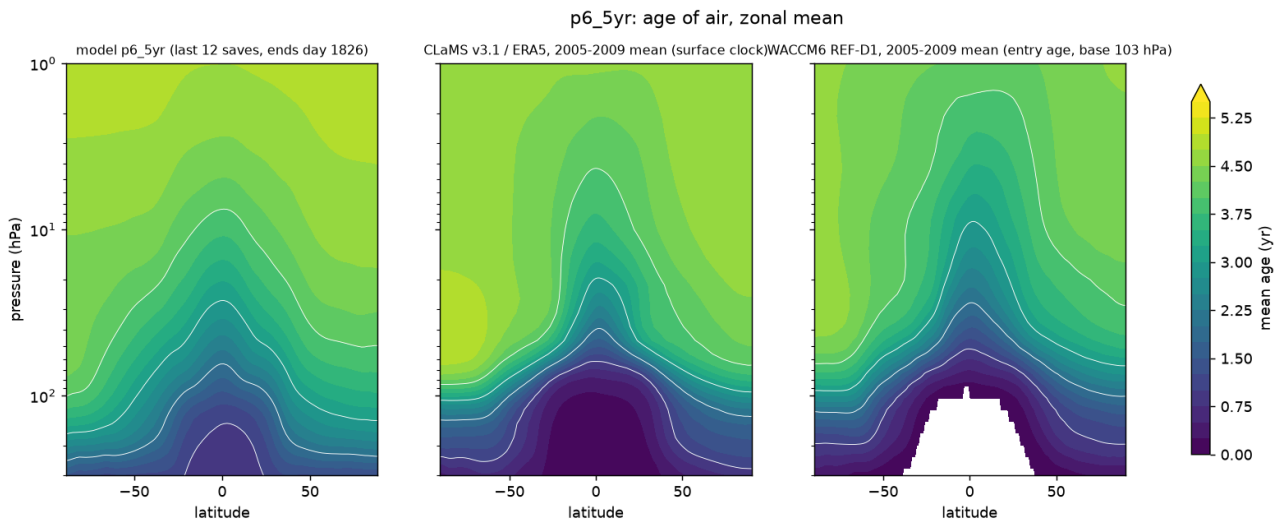


Figure 11. Mean age of air, zonal mean: model (left), CLaMS driven by ERA5 (middle), WACCM6 (right). The pattern is right: young tropical pipe, older extratropics, oldest at the poles. In the lower stratosphere the model's tropics are about a year too old.

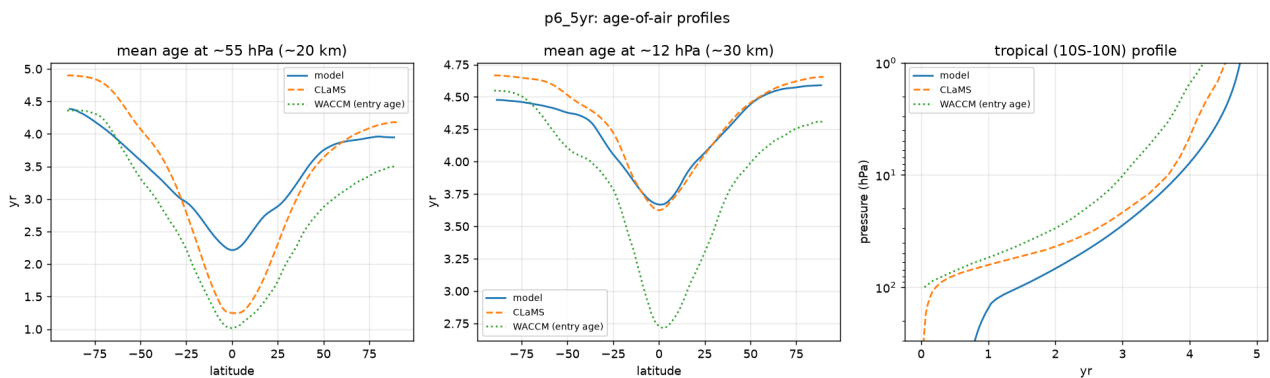


Figure 12. Age of air against latitude at 20 km and 30 km, and the tropical vertical profile. At 30 km the model matches CLaMS almost exactly; at 20 km the tropical minimum is too shallow, which points at the warm tropical tropopause in the standard-atmosphere target.

What remains

- The two missed sudden warmings (2006, 2007). The 15 day relaxation may be too stiff for strongly wave-driven events; a five-year chain at tau 25 would settle it.
- The tropical pipe is too old below 30 hPa. The standard-atmosphere target has no tropical tropopause, so the entry region is 10 K too warm.
- Above 10 hPa the winter pole is now too warm, because the taper hands it back to the standard atmosphere's summer-like values there.
- A bug found on the way: the tracer budget's pull-up check summed over longitude without dividing, so every pull-up ratio printed before 3 September was 192 times too large. No verdict changes, but the Phase 4 record's number should be corrected.

8. Circulation patterns in the five-year chain: QBO, SAO, Brewer-Dobson

diagnostic	model	ERA5 2005-2009	WACCM6	reading
QBO: deseasonalised std of equatorial u at 20 hPa	3.9 m/s	17.5 m/s	10.0 m/s	no QBO; steady easterlies with an annual cycle
SAO amplitude at 1 hPa	5.5 m/s	30.7 m/s	18.4 m/s	present but weak
BDC upward mass flux at 70 hPa, annual	7.2×10^9 kg/s	-	6.1 (same method)	realistic
same, DJF / JJA	8.8 / 6.5	-	8.6 / 5.4	seasonal cycle right
age-of-air transit 70 to 10 hPa (10S-10N)	1.9 yr	CLaMS 2.9 yr	2.5 yr	mid-stratosphere ascent right, entry too old

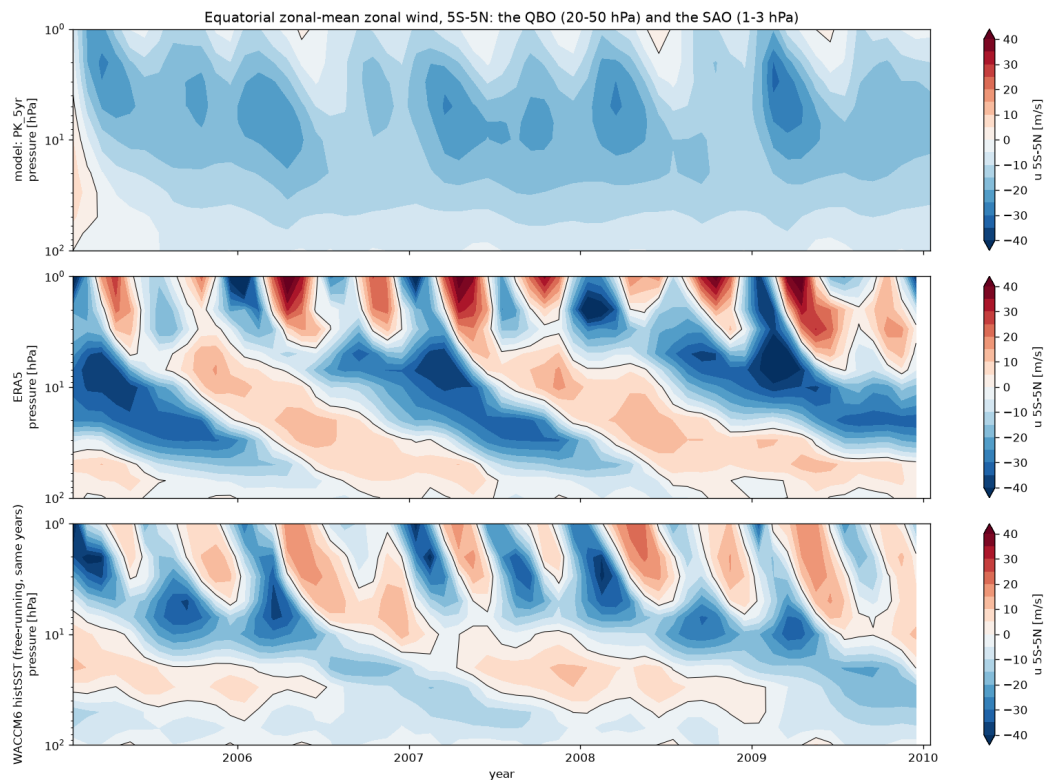


Figure 13. Equatorial zonal wind, 5S to 5N, as a time-height section 2005 to 2009: model (top), ERA5 (middle), WACCM6 (bottom). ERA5 shows the descending westerly and easterly QBO phases between 50 and 10 hPa and the semiannual oscillation near 1 hPa. The model has neither: the equatorial stratosphere stays easterly with only an annual modulation. This was expected: the QBO is driven by tropical waves a dry T63 model without convection or gravity-wave drag does not produce, and the nudging stops at 150 hPa. Nudging the tropical stratospheric wind (issue 6) is the pragmatic fix.

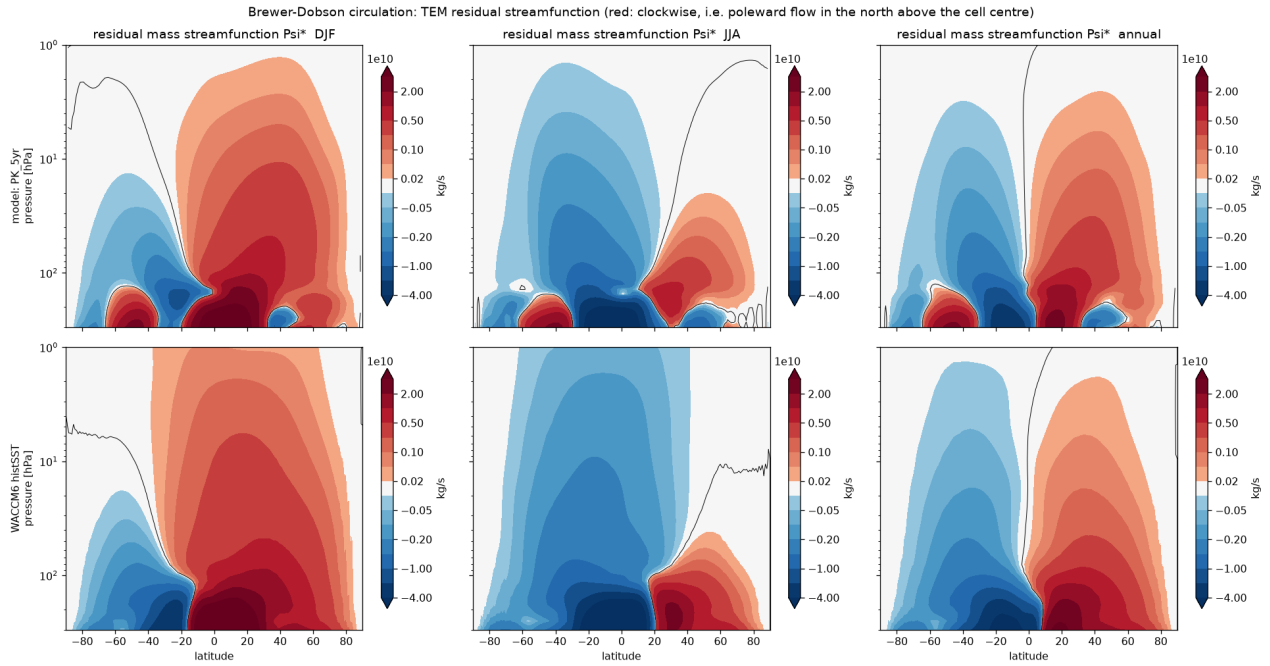


Figure 14. The Brewer-Dobson circulation as the TEM residual mass streamfunction, DJF, JJA and annual: model (top), WACCM6 from its daily zonal-mean TEM tape (bottom). Red is clockwise, so the red cell in DJF carries air up in the tropics and poleward and down in the northern winter hemisphere; blue is the mirror image for the southern winter. Structure, depth and seasonal reversal match, and the tropical upward mass flux at 70 hPa is 7.2 against 6.1×10^9 kg/s. With the troposphere nudged to ERA5 the wave driving is right and the analytic stratosphere lets it act. The model's covariances come from 5-day means, so fast transient waves are missing, which is why its cells are rougher near 100 hPa.

What the ascent tells us

From the age of air, the transit time from 70 to 10 hPa in the tropics is 1.9 years against CLaMS's 2.9, so the mid-stratospheric ascent is if anything slightly fast, yet the air at 70 hPa is already 2.0 years old against 0.75. Too much aged air enters the tropical lower stratosphere, consistent with the warm tropical tropopause of the standard-atmosphere target (issue 31), and is then lifted at roughly the right rate.